YAKEEN

NEET 2.0 2027

Physics | Lecture - 02

Motion in a Plane

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Todays Goal

PhD on Projectile Motion

H/w good

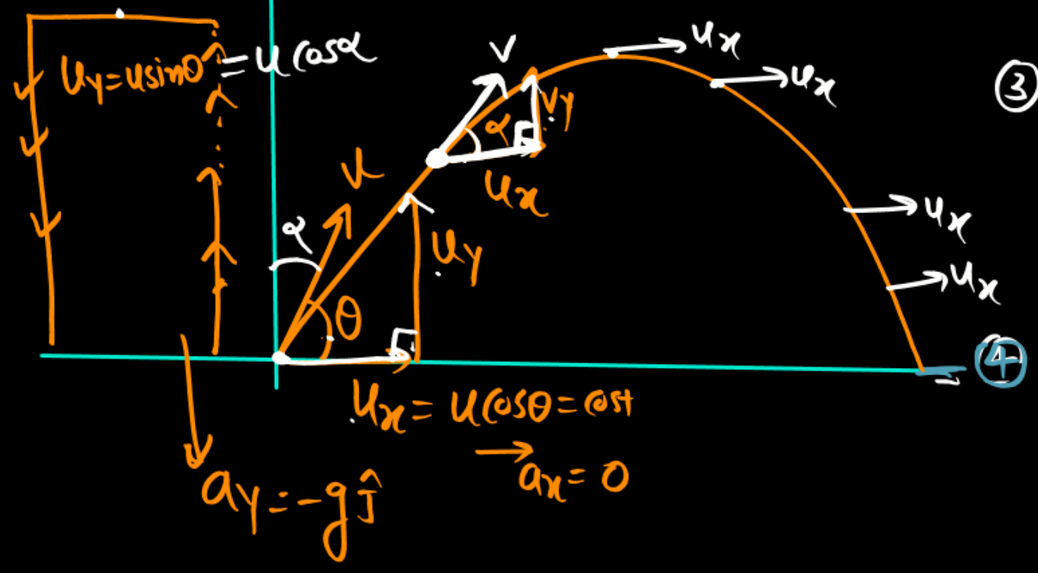
Question in this Notes (PPT) in last attached

Projectile motion → 2-D Non-uniform motion; with uniform acceleration

along y-axis

$v_y = u_y + a_y t$

$\boxed{v_y = u\sin\theta - gt}\ \hat{j}$



① Time of flight $T_f = \frac{2u_y}{g} = \frac{2u\sin\theta}{g}$

$T_f = \frac{2u\cos\alpha}{g}$

② Angle of Projection $\tan\theta = \frac{u_y}{u_x}$

③ $H_{max} = \frac{u_y^2}{2g} = \frac{u^2\sin^2\theta}{2g}$

④ Range (x-axis) $= u_x T_f = u\cos\theta\, T_f$

$= u\cos\theta \times \frac{2u\sin\theta}{g}$

$= \frac{u^2\sin(2\theta)}{g}$

⑤ velocity at max^m Height $= u\cos\theta$

⑥ velocity at time t

$\vec{v_t} = v_{xt}\,\hat{i} + v_{yt}\,\hat{j}$

$= u\cos\theta\,\hat{i} + (u\sin\theta - gt)\,\hat{j}$

speed $= \sqrt{(u\cos\theta)^2 + (u\sin\theta - gt)^2}$

⑦ Direction of Motion at time 't'.

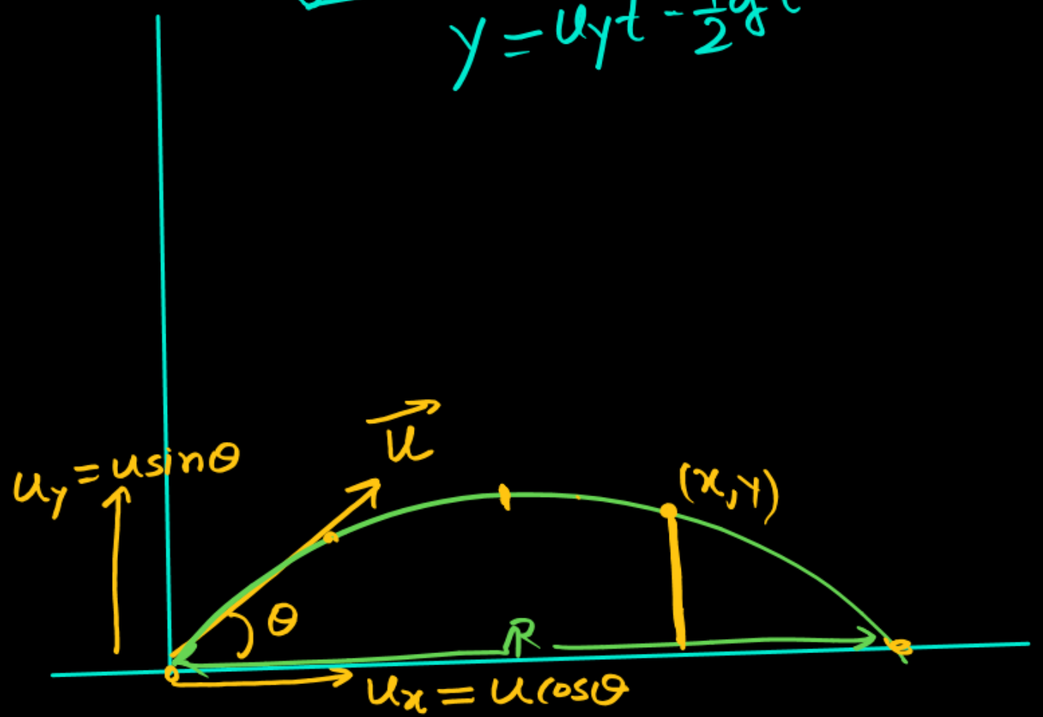
$\tan\alpha = \frac{v_y}{u_x} = \frac{u\sin\theta - gt}{u\cos\theta}$

↑ from x-axis

Projectile motion → 2-D ..... motion with ..... acceleration

for y-axis

$$y = u_y t - \frac{1}{2} g t^2$$



⑧ Position at time 't'

$$\vec{r} = x\hat{i} + y\hat{j}$$

$$x = u_x t = (u\cos\theta\, t)\hat{i} \quad -①$$

$$y = u_y t - \frac{1}{2} g t^2$$

$$y = (u\sin\theta\, t - \frac{1}{2} g t^2)\hat{j} \quad -⑪$$

⑨ eqn of Trajectory

Relⁿ b/w x & y

Put t in eqn ⑪

$$y = u\sin\theta\left[\frac{x}{u\cos\theta}\right] - \frac{1}{2} g \left(\frac{x}{u\cos\theta}\right)^2$$

$$y = x\tan\theta\left[1 - \frac{gx}{2u^2\cos^2\theta\sin\theta/\cos\theta\cos\theta}\right]$$

$$y = \tan\theta\, x - \frac{g x^2}{2u^2\cos^2\theta}$$ → Parabolic

Radia

$$y = x\tan\theta\left[1 - \frac{x}{R}\right]$$

⑩ Relation b/w H_{max} & Range(R)

$$H_{max} = \frac{u_y^2}{2g} \qquad R = u_x T_f = u_x \frac{2u_y}{g}$$

$$\frac{H_{max}}{R} = \frac{u_y^2}{2g} \cdot \frac{u_y}{2u_x} \cdot g$$

$$\frac{H_{max}}{R} = \frac{u_y}{4u_x}$$

$$\frac{H_{max}}{R} = \frac{\tan\theta}{4}$$

$$\boxed{H_{max} = \frac{R\tan\theta}{4}}$$

(11) Avg velocity in compt Journey

$$\text{Avg velocity} = \frac{R}{T_f} = \frac{u_x T_f}{T_f}$$

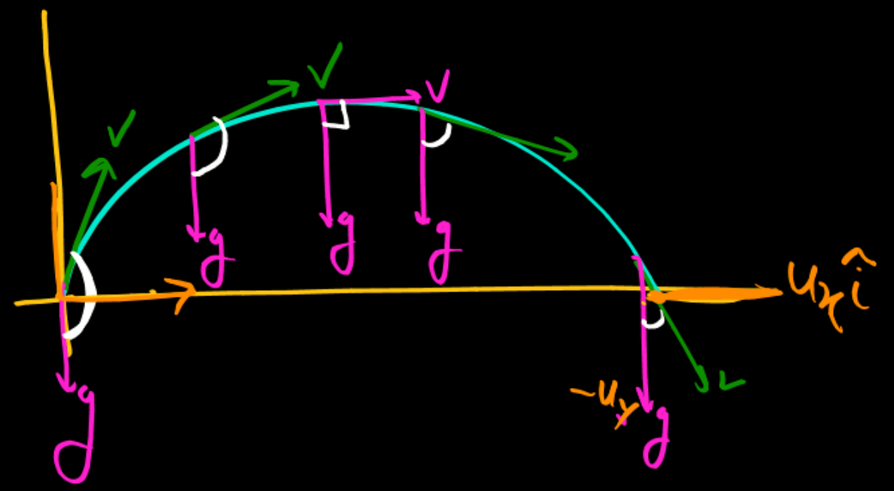
$$= u_x = u\cos\theta$$

✓✓ $$\text{Avg velocity} = \frac{\vec{u_i} + \vec{v_f}}{2} = u\cos\theta$$

(12) Angle b/w velocity & accn.

→ continously decreasing

at max height $\vec{v}$ & $\vec{a}$



(13) change in velocity in Projectile

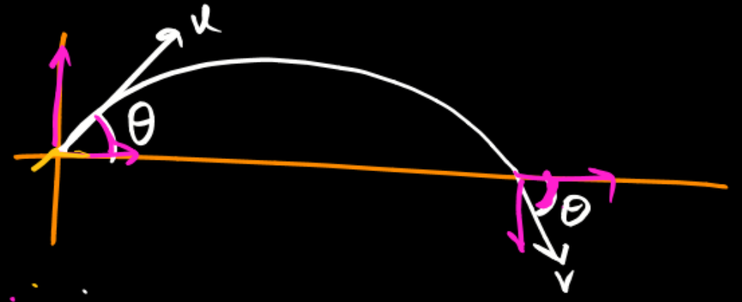
$$\vec{\Delta v} = \vec{v_f} - \vec{v_i} = -2u_y$$

$$= -2u\sin\theta$$ ✓

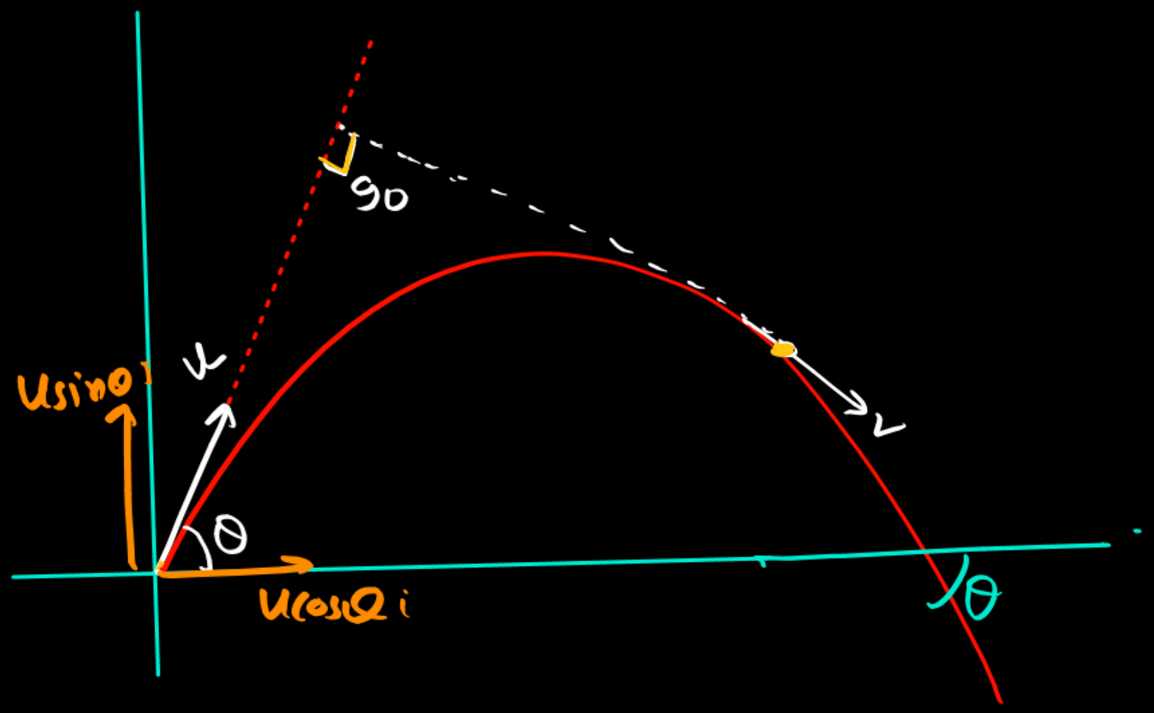
(14) final velocity of collison.

$$\vec{u}_{proje} = u_x\hat{i} + u_y\hat{j} \qquad \vec{v_f} = u_x\hat{i} - u_y\hat{j}$$

(15) Total Angular displacement by Projectile in compt Journey. $= 2\theta$ ✓



(16) find time after which Velocity is perpendicular to intial direction of Projection.



$$\begin{cases} \vec{u} = u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j} \\ \vec{v_t} = u\cos\theta\,\hat{i} + (u\sin\theta - gt)\,\hat{j} \end{cases}$$

Angle b/w $\vec{v_t}$ & $\vec{u_i}$ is 90°.

$$\vec{v_t} \cdot \vec{u_i} = 0$$

$$[u\cos\theta\, i + (u\sin\theta - gt)\hat{j}] \cdot [u\cos\theta\, i + u\sin\theta\, \hat{j}] = 0$$

$$u^2\cos^2\theta + u^2\sin^2\theta - gtu\sin\theta = 0$$

$$u^2[\sin^2\theta + \cos^2\theta] = gtu\sin\theta$$

$$u\,u^2 = gtu\sin\theta$$

$$\boxed{t = \frac{u}{g\sin\theta}}$$

(17) Rate of change in momentum in 1-sec

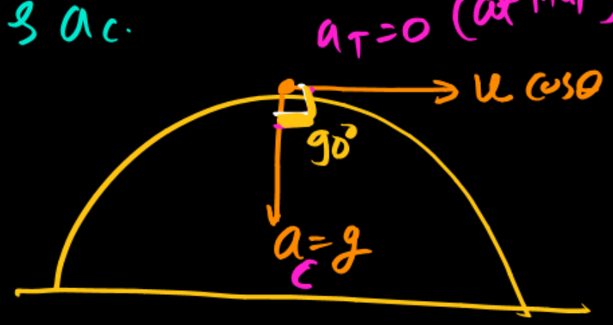
Soln

Force = The rate of change in momentum

Force = mg (downward) Ans

(18) acc^n at Maximum Height

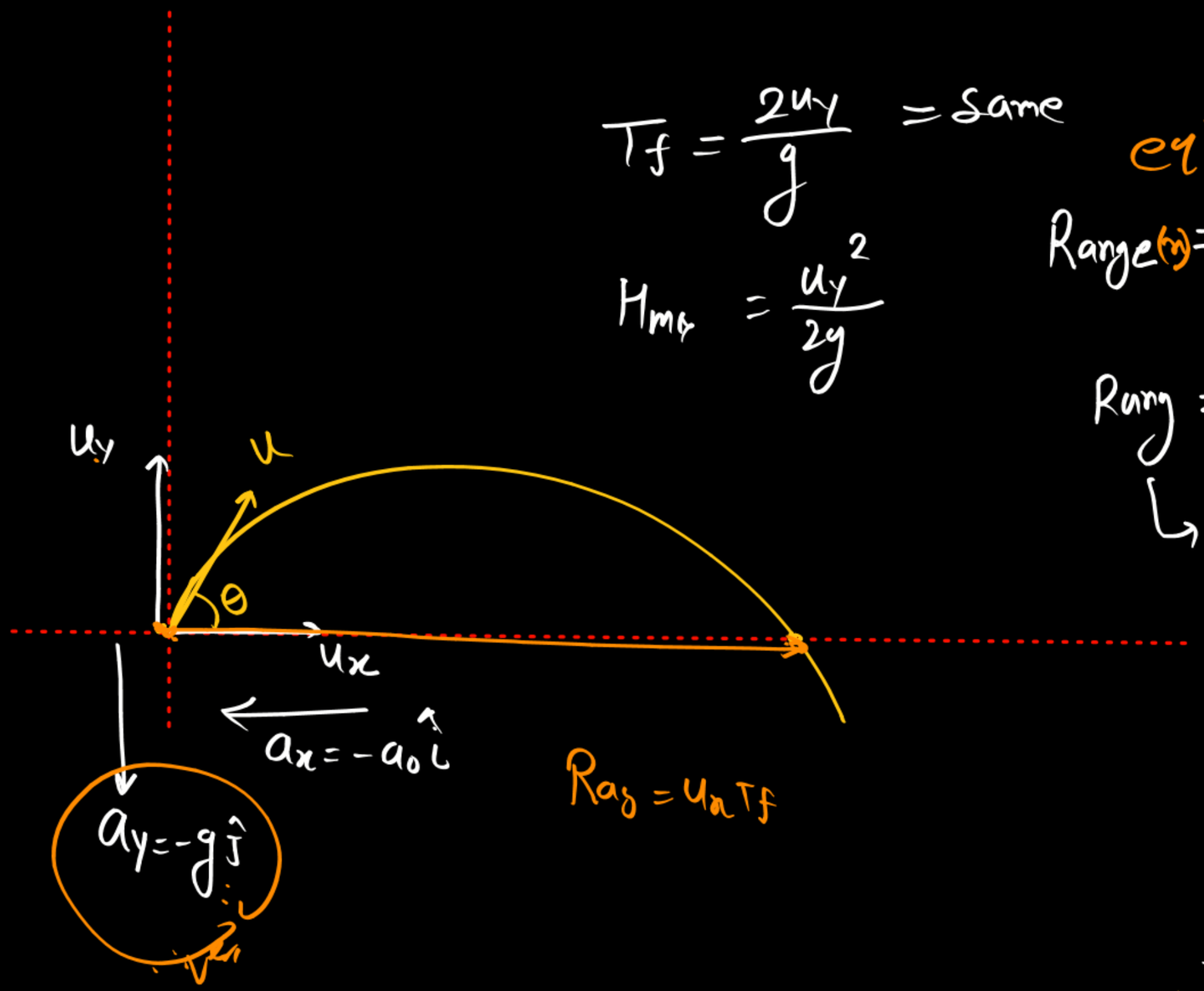
a_T & a_c.



(19) acc^n at $t = 4$ sec.

$$\vec{a} = -g\hat{j} + 0\hat{i}$$

(20) If constant air friction act along -ve x-axis a_0 then find H, T & R.



$$T_f = \frac{2u_y}{g} = \text{same}$$

$$H_{max} = \frac{u_y^2}{2g}$$

$$R_{ang} = u_x T_f$$

eqⁿ of motⁿ in x-axis

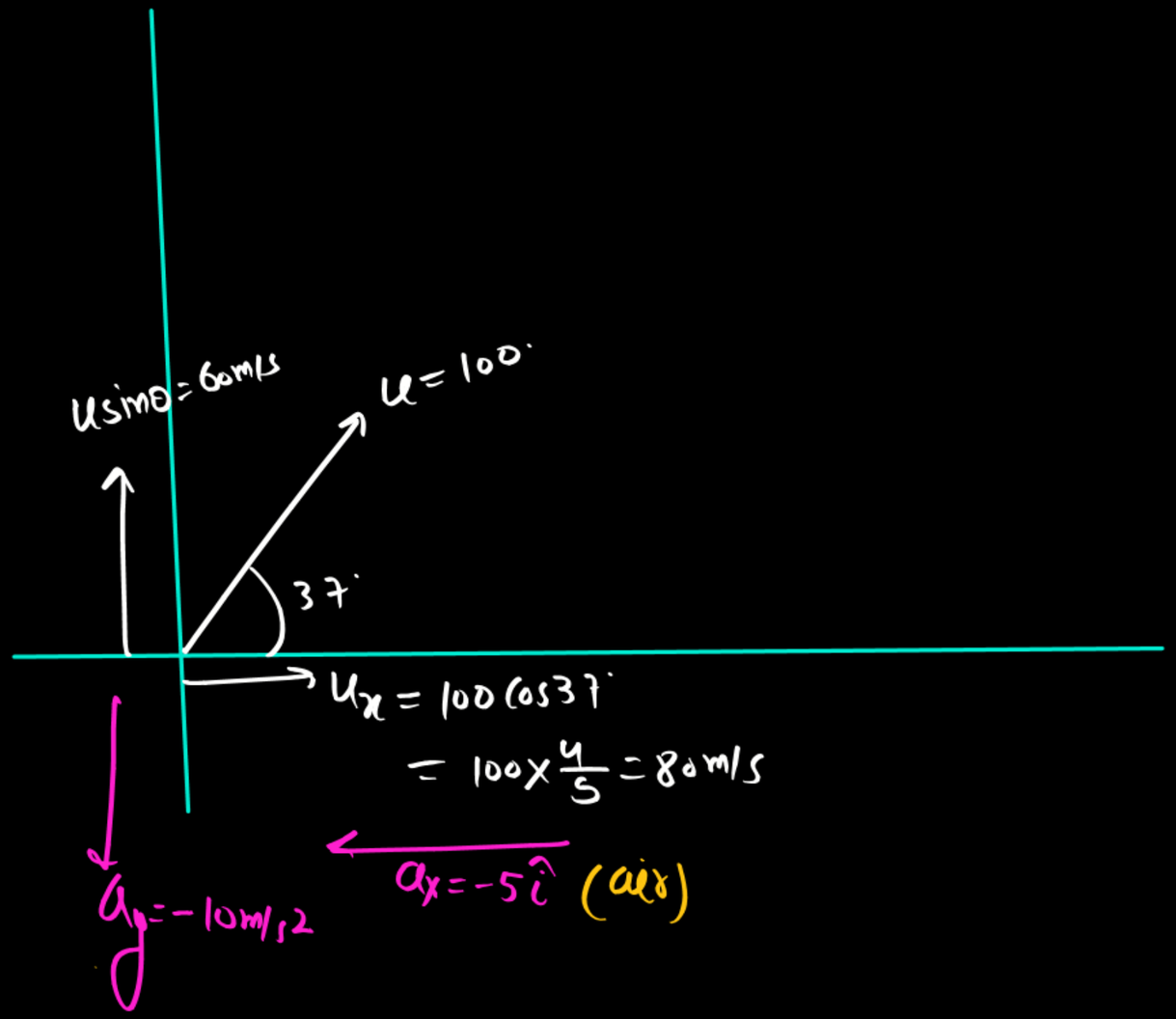
$$\text{Range}(x) = u_x t + \frac{1}{2} a_x t^2$$

$$R_{ang} = u_x(t_f) - \frac{1}{2} a_0 T_f^2 \quad (\text{Put } t_f)$$

find Rang.

↳ change ✓

(21) 9 f object is Projected 100 m/s at 37° then f H, T, R if acc^{n} due to air friction $\vec{a}_{air} = -5\hat{i}$



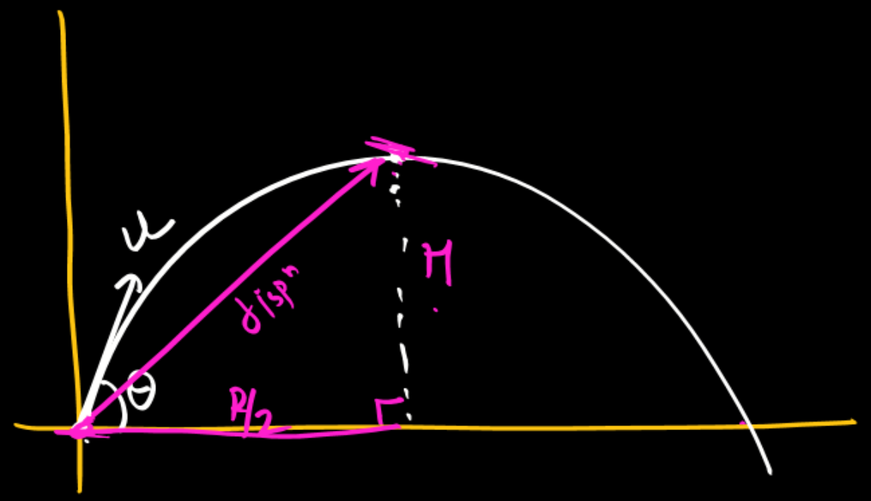
$$T_f = \frac{2u_y}{g} = \frac{2\times 60}{10} = 12\ \text{sec}$$

$$H_{max} = \frac{u_y^2}{2g}$$

$$R_{net}(x) = u_x T_f - \frac{1}{2} a_x T_f^2$$

$$x = 80 \times 12 - \frac{1}{2}(5)(12)^2$$

(22) Avg velocity b/w Point of Projection to maxm height



$\vec{u}_{Avg} = u\cos\theta$ ← cosθ Jouray : ka

$$\text{Avg velocity} = \frac{\sqrt{H^2 + (R/2)^2}}{\left[\frac{u_y}{g}\right]}$$

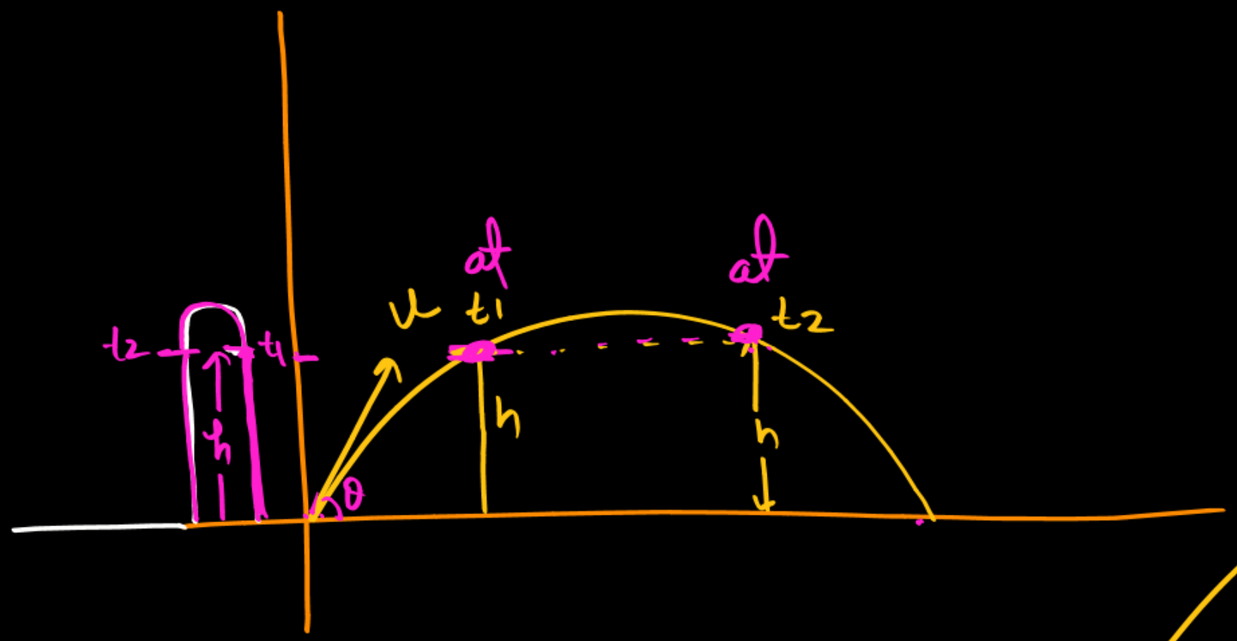
$$u_{Avg} \text{ (Point of Proj to maxm hey)} = \frac{\vec{u_i} + \vec{v_f}}{2}$$

$$= \frac{(u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}) + u\cos\theta\,\hat{i}}{2}$$

Same

$$\text{Avg } \vec{v} = \frac{2u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}}{2}$$

(23) gf Particle is at same height at time t_1 & t_2 then find T_f, H_{max} & that height.



$$u = \frac{g(t_1+t_2)}{2\sin\theta} \checkmark$$

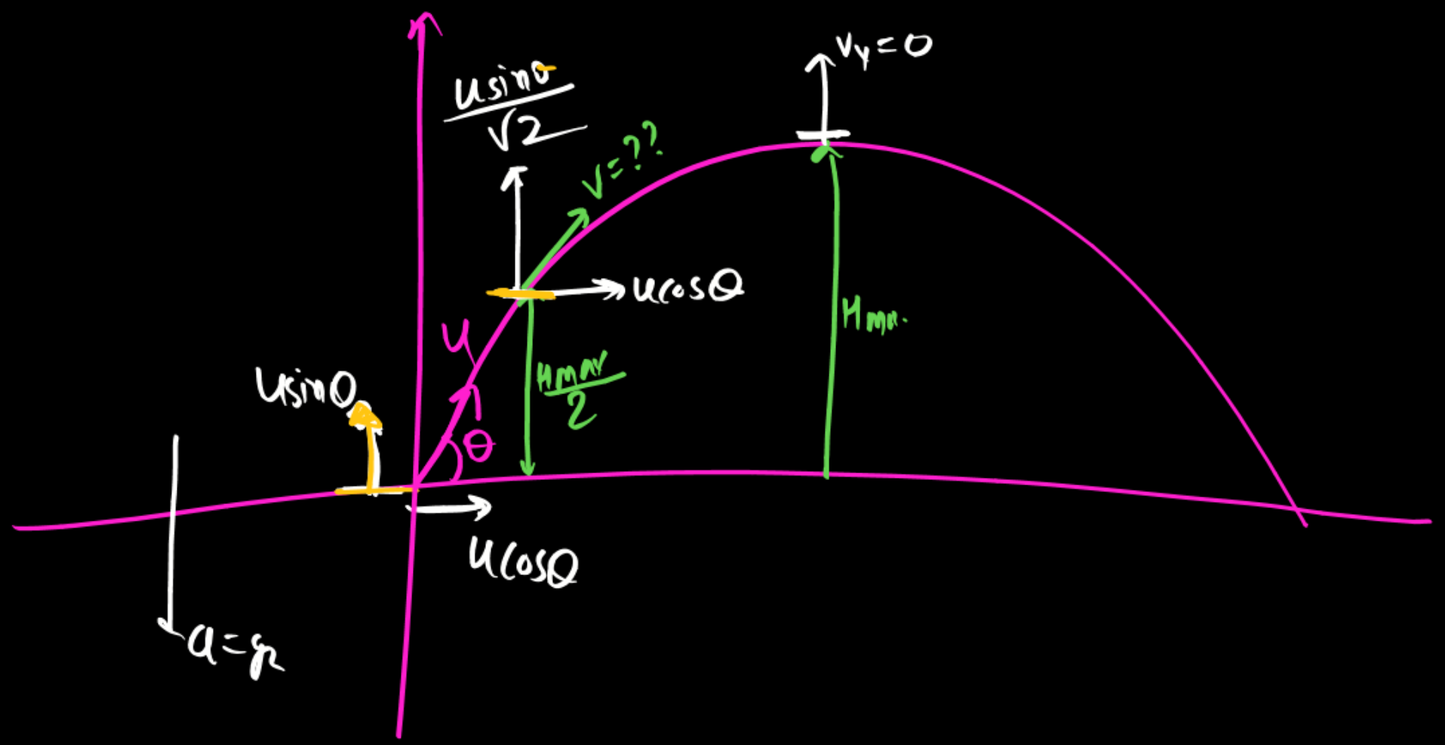
$$T_f = t_1+t_2 = \frac{2u\sin\theta}{g}$$

$$h = \frac{1}{2} g t_1 t_2$$

Not a H_{max}

(29) Find Velocity of Particle at Half of maximum Height.

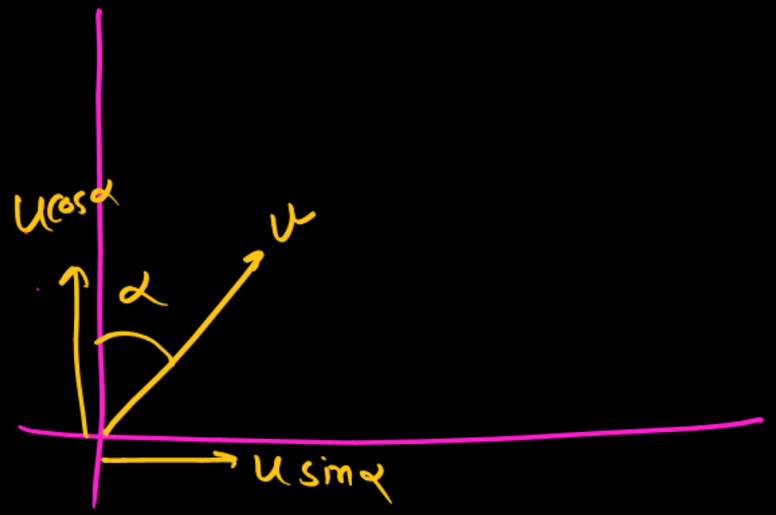
Object is project with speed u at angle θ.



$$\vec{V} = u\cos\theta\,\hat{i} + \frac{u\sin\theta}{\sqrt{2}}\,\hat{j}$$

$$\text{speed} = \sqrt{(u\cos\theta)^2 + \left(\frac{u\sin\theta}{\sqrt{2}}\right)^2}$$

(25) If Ball is projected from vertical at angle α then find H, T & R.



$T_f = \frac{2u_y}{g} = \frac{2u\cos\alpha}{g}$

$H_{max} = \frac{u^2\cos^2\alpha}{2g}$

$R = u_x T_f = \frac{u\sin\alpha \; 2u\cos\alpha}{g} = \frac{u^2 \sin 2\alpha}{g}$ ✓

$= u_x \frac{2u_y}{g}$

(26) what should be θ so that Range will be maximum :—

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$R_{max} = \frac{u^2 [\sin 2\theta]_{max}}{g}$$

$$R_{max} = \frac{u^2 \, 1}{g}$$

$2\theta = 90^\circ$

$\theta = 45^\circ$

$$R_m = \frac{u^2}{g}$$

$R_{44} = R_{46}$ ⟶ yes

$R_{45-\theta} = R_{45+\theta}$ ⟶ yes

$R_\alpha = R_\beta$ ⟶ yes

$\alpha + \beta$ must be 90°

↳ (complementy angle)

(17) Two Ball is projected with same speed u at angle α and β. if $\alpha + \beta = 90^\circ$ then Range will be same Complementray Angle.

α	β
30°	60°
37°	53°
10°	80°
25°	65°
55°	35°

Range

(28) With same speed Ball is proted at angle.

43° & 48°

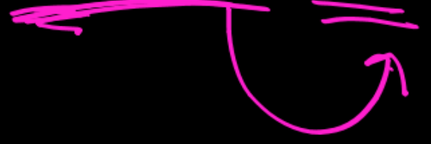
(a) $R_{43} = R_{48}$

(b) $R_{43} > R_{48}$ (62°)

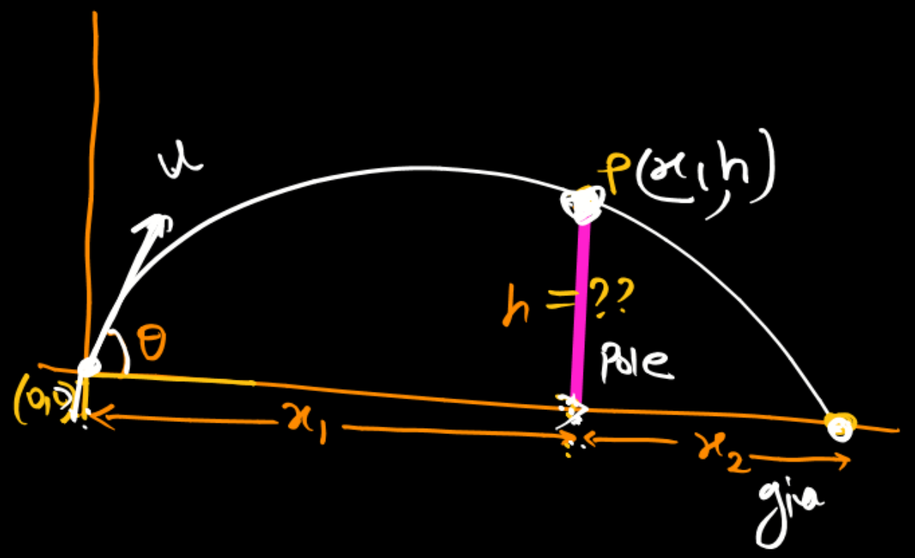
(c) $R_{43} < R_{48}$

R_{45°

$R_{43} = R_{47} > R_{48}$



(2) find height??



use eqn of trajectory,

$$y = x\tan\theta\left[1-\frac{x}{R}\right]$$

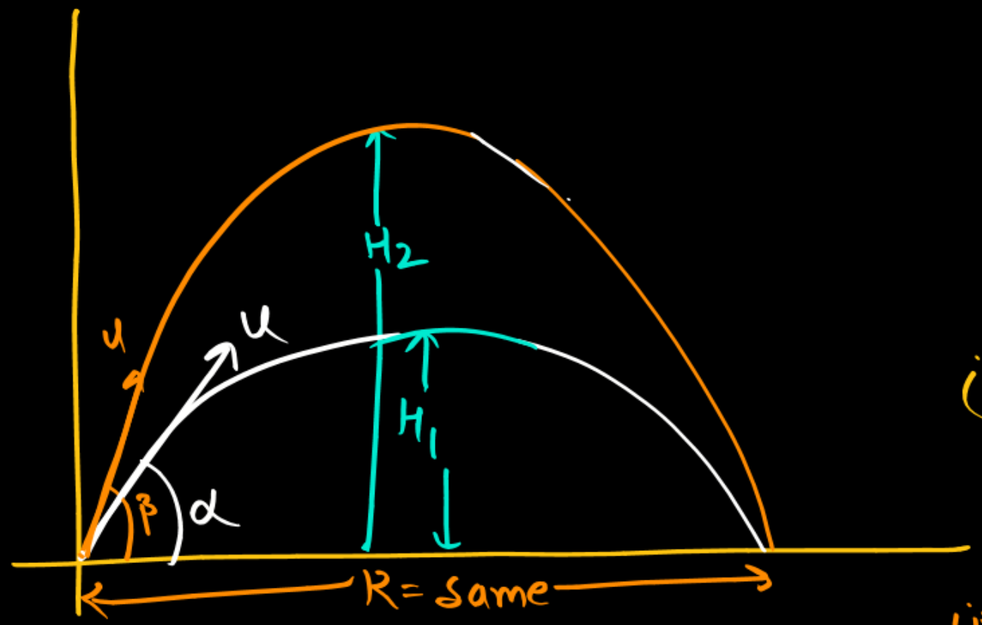
$$h = x_1\tan\theta\left[1-\frac{x_1}{x_1+x_2}\right]$$

$$= x_1\tan\theta\left[\frac{x_2}{x_1+x_2}\right]$$

$$h = \frac{x_1 x_2\tan\theta}{x_1+x_2}$$

(3) Two Ball is Projected with same speed at angle α & β such that $\alpha+\beta=90^\circ$ (IIT-ADV 2016)

then find $\left(\frac{H_1}{H_2}\right)_{max\,ratio}$.



(i) $\frac{H_1}{H_2} = \frac{\frac{u_{y_1}^2}{2g}}{\frac{u_{y_2}^2}{2g}}$

$\frac{H_1}{H_2} = \frac{u^2\sin^2\alpha}{u^2\sin^2\beta} = \frac{\sin^2\alpha}{\sin^2(90-\alpha)} = \frac{\sin^2\alpha}{\cos^2\alpha} = \tan^2\alpha$

(ii) $\frac{T_1 f}{T_2 f} = \frac{\frac{2u\sin\alpha}{g}}{\frac{2u\sin\beta}{g}} = \frac{\sin\alpha}{\sin(90-\alpha)} = \frac{\sin\alpha}{\cos\alpha} = \tan\alpha$

(iii) $H_1 \cdot H_2 = ??$

$H_1 H_2 = \frac{R\tan\alpha}{4}\,\frac{R\tan\beta}{4}$

$H_1 H_2 = \frac{R^2}{16}\tan\alpha\tan(90-\alpha)$

$H_1 H_2 = \frac{R^2}{16}\tan\alpha\cot\alpha$

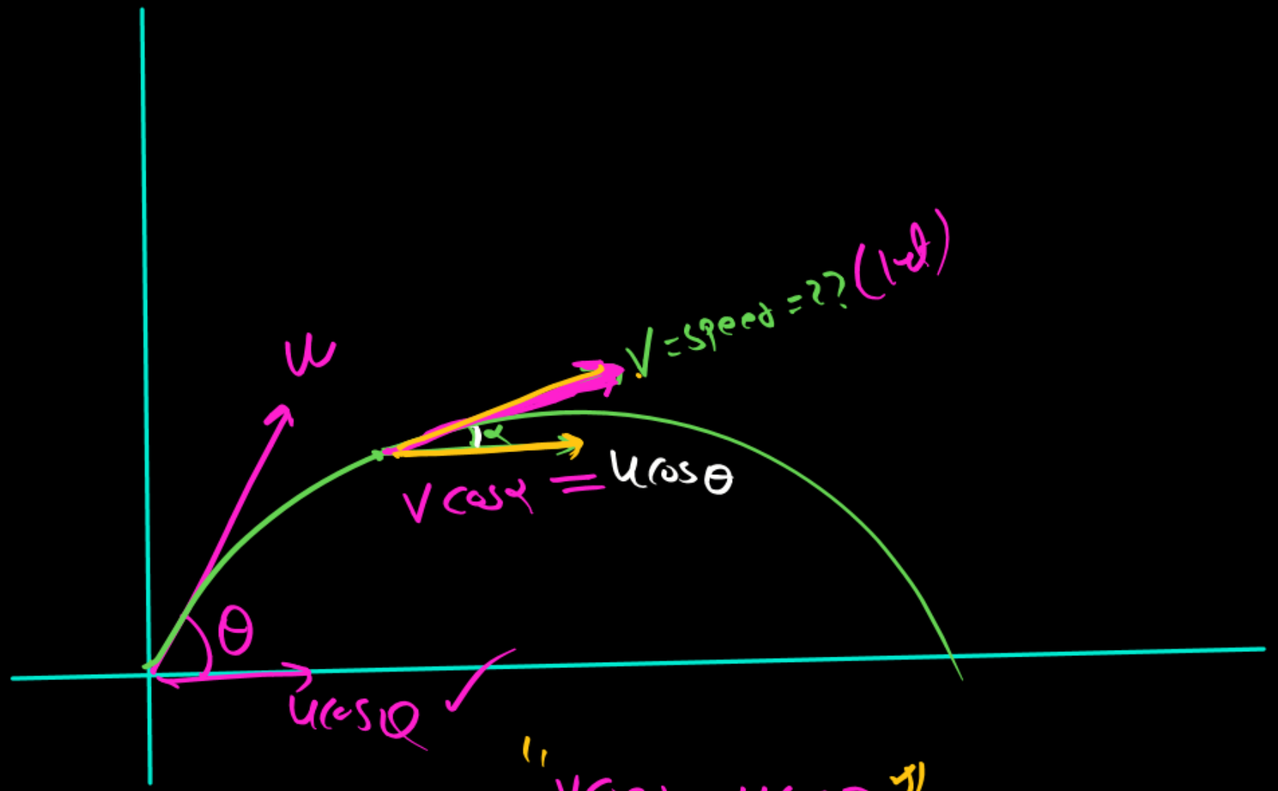
$H_1 H_2 = \frac{R^2}{16}$

* $H = \frac{R\tan\theta}{4}$

$\sin(90-\alpha) = \cos\alpha$

$\tan(90-\alpha) = \cot\alpha$

(31) object is Projected with speed u at angle θ then find its speed when object is moving at angle α from horizontal

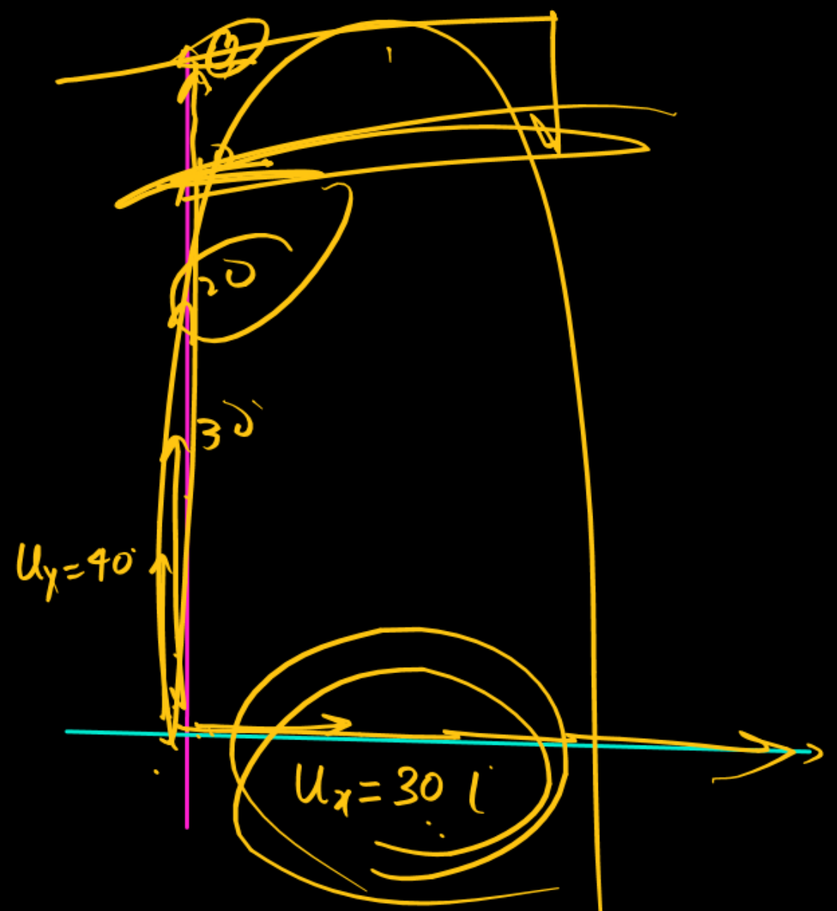


" V cosα = u cosθ "

$$V = \frac{u\cos\theta}{\cos\alpha} = u\cos\theta \cdot \sec\alpha$$

Copy all 30 Question

Object is Projected with $\vec{u} = 30\hat{i} + 40\hat{j}$ then find



(1) speed of Projection $=$

$$\sqrt{(30)^2 + (40)^2}$$

$$= 50\ m/s$$

② Angle of Projection

$$\tan\alpha = \frac{40}{30}$$

$$\alpha = 53^\circ$$

③ $T_f = 4 + 4 = 8$.

④ $H_{max} = 80\ m$

⑤ Range $= 30 \times 8 = 240\ m$

⑥ velocity after 2-sec

$$v_x = 30i + 20i$$

⑦ Positiat after (3-sec)

$$x = 90i + 75j$$

⑧ velocity at 4-sec

$$\vec{V} = 30 + 0\hat{j}$$

⑨ Position at time t

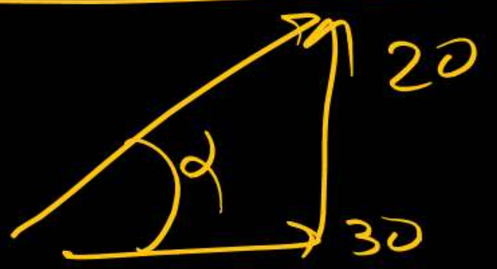
$$x = 30t$$

$$y = 40t - \frac{1}{2}10t^2$$

⑩ Equation of Trajectory

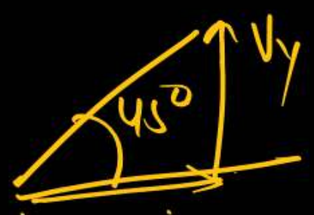
$$y = 40\left(\frac{x}{30}\right) - \frac{1}{2}10\left(\frac{x}{30}\right)^2$$

(11) dirn of motion at $t = 2\,sec$



$$\tan\alpha = \frac{20}{30}$$

(12) find time when object is moving 45° from x-axis



$$\tan 45 = \frac{V_y}{V_x}$$

$$\tan 45 = \frac{40 - 10t}{30}$$

$$1 = \frac{40 - 10t}{30}$$

(13) Velocity of Collision

$$V_f = 30i - 40j$$

(14) find time when velocity is $\perp$ to intial direction of motion.

$$t = \frac{u}{g\sin\theta} = \frac{50}{10\times\sin\theta}$$

(15) velocity of collsion.

(16) change in velocity

$$\vec{\Delta V} = -80$$

(17) Avg velocity in on

$$= v_{av} = 30 \text{ m/s}$$

(18) If $m = 4\,kg$ momentum at 3sec.

$$\vec{p} = m\vec{v}$$

$$= 4\left[30\hat{i} + 10\hat{j}\right]$$

(19) Rate of change in Momentum at $t = 1\,sec$

$$F = ma$$

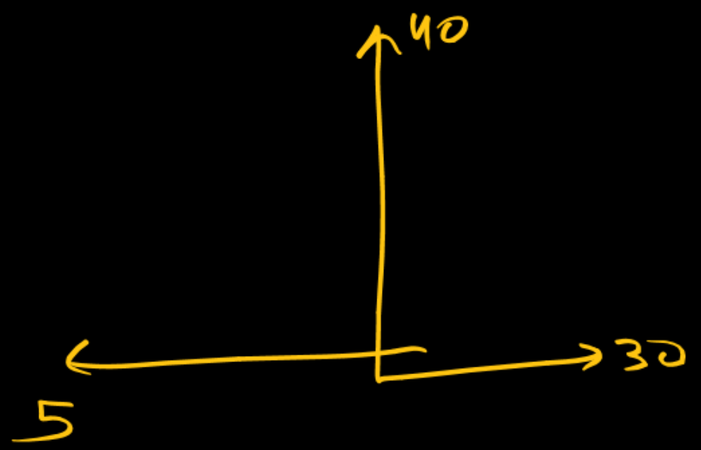
(20) find acc^n at $t = 5.7\,sec$

$$a = g$$

(21) gf Const^n air friction produce acc^n $a = -5\,m/s^2\,\hat{i}$ then find H/T/R.

$T_f = 8$

$l_{tr} = 80\,m$



$$R_{ay} = 30 \times 8 - \frac{1}{2}(5)(8)^2$$

$$= (240 - 5 \times 32)$$

Copy (21) question in note

Home - work

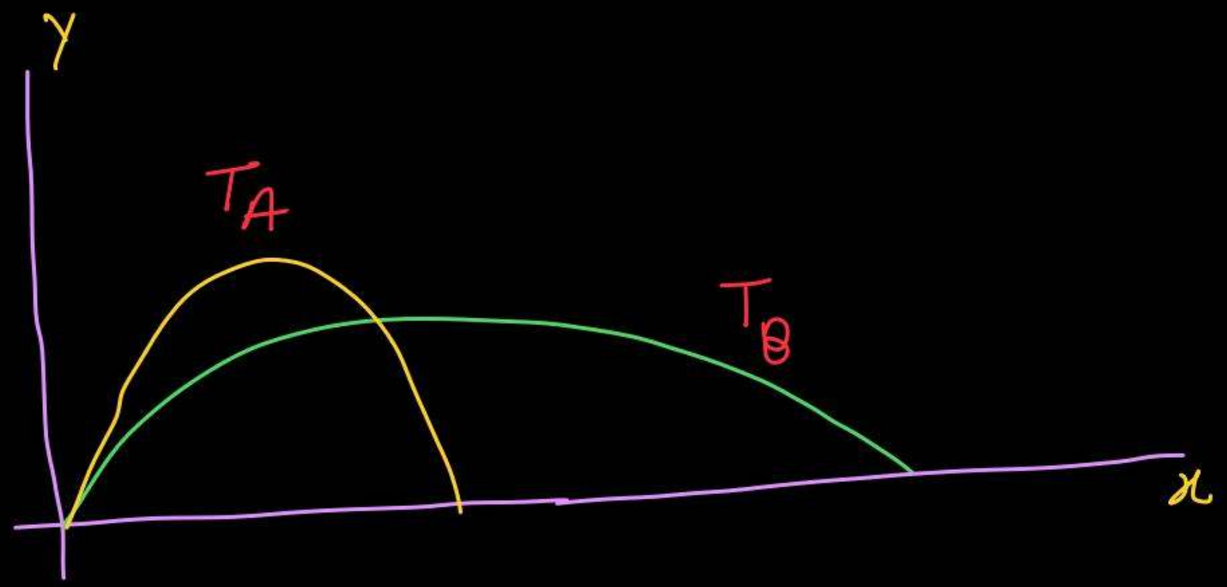
Sare karne hai Question

Must for all

Which of the following physical quantity remains constant in projectile Motion.

(a) speed (b) velocity (c) acceleration (d) Horizontal velocity

(e) tangential acceleration.



Which of the following is correct :-

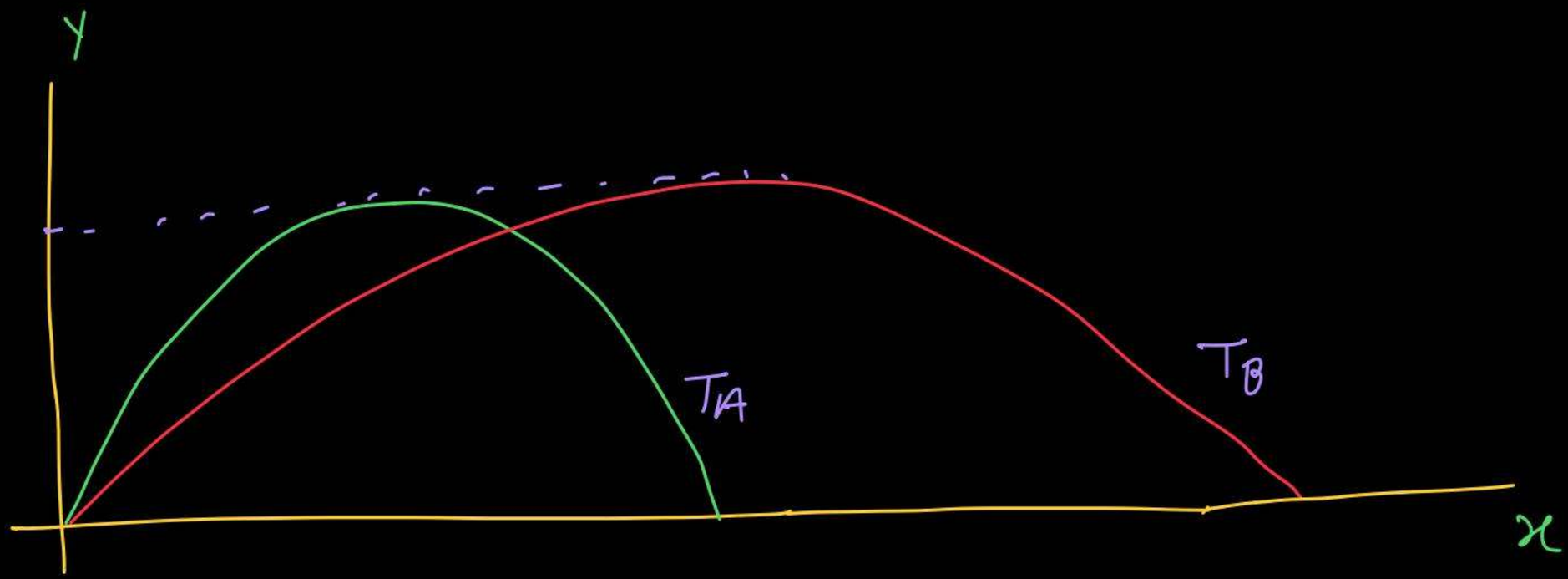
(a) $T_A = T_B$

(b) $T_A > T_B$

(c) $T_A < T_B$

(26)

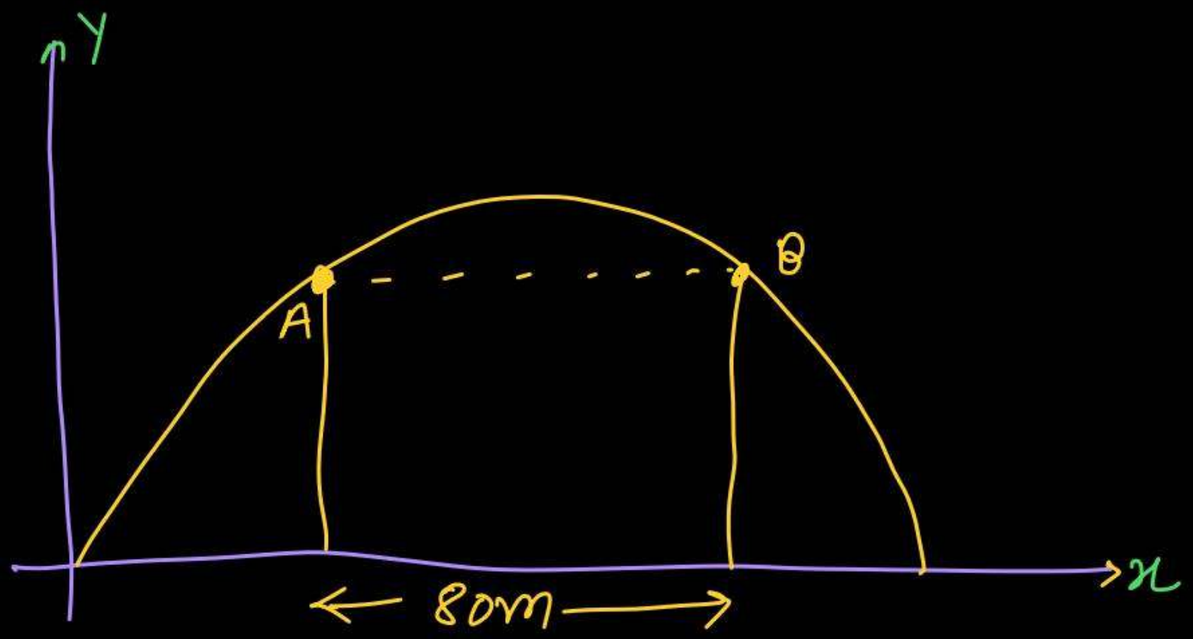
Ball is projected with u at diff^{n} Angle many times then find maximum Area can be covered by Projectile.



$T_A = T_B$

$T_A > T_B$

$T_A < T_B$



Ball is at same height
$t_A = 4sec$ & $t_B = 8sec$ then
find speed of Projection
& Angle of Projection.

Ball is projected at two diffrent angle such that range are same & time of flight $t_1 = 4sec$ & $t_2 = 6sec$. then find Horizontal Range. (IIT)

Ball is Projected and its Position at time t is $\vec{r} = 30t\,\hat{i} + (40t - 5t^2)\,\hat{j}$

Then find Range :-

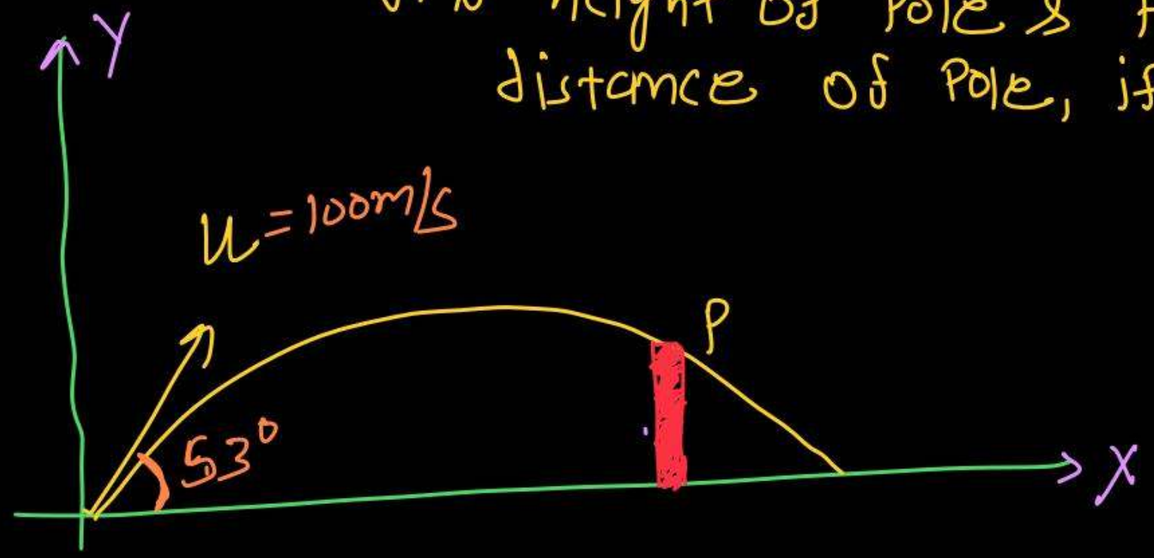
Solution

Ball is projected with 100m/s at 53° then find its speed when it is moving at 37° from Horizontal. (H|W)

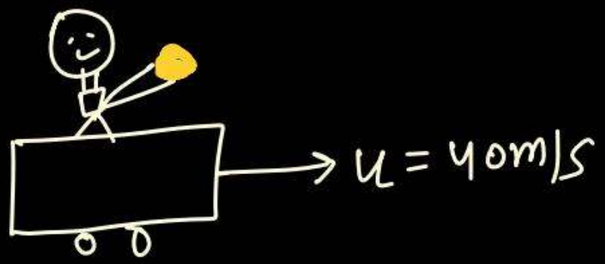
(8) Ball is projected with speed 100m/s at 53° then find time when its velocity parallel to x-axis and velocity 37° from x axis.

Ball is projected at angle 45° and it touches top of pole at dist^{n} 5m and fall other side 2m from pole then find height of pole.

find height of Pole & Horizontal distance of Pole, if Ball touches the Pole at $t = 2\,sec$



9f Ramlal Project Ball vertically up with velocity 30m/s then find H_{max} & Range of Ball



Question

For angles of projection of a projectile $(45^\circ - \theta)$ and $(45^\circ + \theta)$, the horizontal ranges described by the projectile are in the ratio of: -

AIPMT 2006 (+4 /-1)

1. 1 : 1
2. 2 : 3
3. 1 : 2
4. 2 : 1

Question

A bus is moving with a speed of 10 m/s on a straight road. A scooterist wishes to overtake the bus in 100 s . If the bus is at a distance of 1 km from the scooterist, with what speed should the scooterist chase the bus? **AIPMT 2009 (+4 /-1)**

1. 10 m/s
2. 20 m/s
3. 40 m/s
4. 25 m/s

Question

The speed of a projectile at its maximum height is half of its initial speed. The angle of projection is: -

AIPMT (Pre) 2010 (+4 /-1)

1. 15°
2. 30°
3. 45°
4. 60°

Question

A missile is fired for maximum range with an initial velocity of 20 m/s.

If $g = 10\ m/s^2$, the range of the missile is: - **AIPMT (Pre) 2011 (+4 /-1)**

1. 40 m
2. 50 m
3. 60 m
4. 20 m

Question

A projectile is fired at an angle of 45° with the horizontal. Elevation angle of the projectile at its highest point as seen from the point of projection, is:

AIPMT (Mains) 2011 (+4 /-1)

1. $45°$
2. $60°$
3. $\tan^{-1}\left(\frac{1}{2}\right)$
4. $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

Question

The horizontal range and the maximum height of a projectile are equal. The angle of projection of the projectile is: -

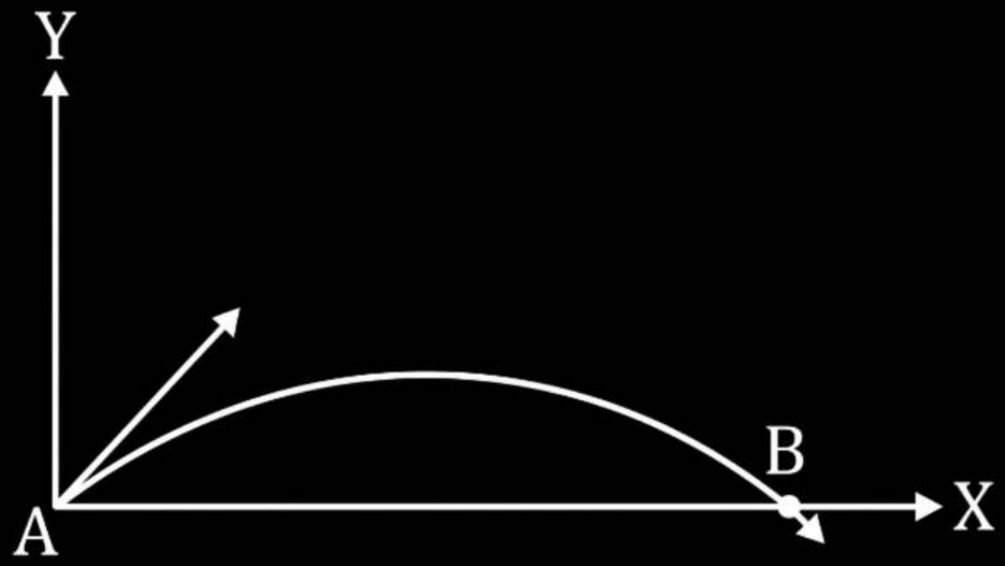
AIPMT (Pre) 2012 (+4 /-1)

1. $\theta = \tan^{-1}(2)$
2. $\theta = 45^\circ$
3. $\theta = \tan^{-1}\left(\frac{1}{4}\right)$
4. $\theta = \tan^{-1}(4)$

Question

The velocity of a projectile at the initial point A is $(2\hat{i} + 3\hat{j})$ m/s. Its velocity (in m/s) at point B is: -

NEET-UG 2013 (+4 /-1)



1. $2\hat{i} + 3\hat{j}$
2. $-2\hat{i} - 3\hat{j}$
3. $-2\hat{i} + 3\hat{j}$
4. $2\hat{i} - 3\hat{j}$

Question

A projectile is fired from the surface of the earth with a velocity of 5 m/s and angle θ with the horizontal. Another projectile fired from another planet with a velocity of 3 m/s at the same angle follows a trajectory which is identical with the trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in m/s^2) is: (given $g = 9.8\ m/s^2$) **AIPMT 2014 (+4 /-1)**

1. 3.5
2. 5.9
3. 16.3
4. 110.8

Question

A ship A is moving Westwards with a speed of 10 km/h and a ship B 100 km South of A, is moving Northwards with a speed of 10 km/h. The time after which the distance between them becomes shortest, is: -

AIPMT 2015 (+4 /-1)

1. 5 h
2. $5\sqrt{2}$ h
3. $10\sqrt{2}$ h
4. 0 h

Question

Two particles A and B, move with constant velocities $\vec{v}_1$ and $\vec{v}_2$. At the initial moment their position vectors are $\vec{r}_1$ and $\vec{r}_2$ respectively. The condition for particle A and B for their collision is: -

Re-AIPMT 2015 (+4 /-1)

1. $\vec{r}_1 - \vec{r}_2 = \vec{v}_1 - \vec{v}_2$
2. $\frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = \frac{\vec{v}_2 - \vec{v}_1}{|\vec{v}_2 - \vec{v}_1|}$
3. $\vec{r}_1 \cdot \vec{v}_1 = \vec{r}_2 \cdot \vec{v}_2$
4. $\vec{r}_1 \times \vec{v}_1 = \vec{r}_2 \times \vec{v}_2$

Question

The speed of a swimmer in still water is 20 m/s. The speed of river water is 10 m/s and is flowing due east. If he is standing on the south bank and wishes to cross the river along the shortest path, the angle at which he should make his strokes w.r.t. north is given by:

NEET(UG) 2019 (+4 /-1)

1. 30° west
2. 0°
3. 60° west
4. 45° west

Question

When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object the shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be

NEET(UG) 2019 (+4 /-1)

1. $1:\sqrt{2}$
2. $\sqrt{2}:1$
3. $1:\sqrt{3}$
4. $1:2\sqrt{3}$

Question

A person standing on the floor of an elevator drops a coin. The coin reaches the floor in time t_1 if the elevator is at rest and in time t_2 if the elevator is moving uniformly. Then: -

NEET(UG) 2019 (Odisha) (+4 /-1)

1. $t_1 < t_2$ or $t_1 > t_2$ depending upon whether the lift is going up or down
2. $t_1 < t_2$
3. $t_1 > t_2$
4. $t_1 = t_2$

Question

Two bullets are fired horizontally and simultaneously towards each other from roof tops of two buildings 100 m apart and of same height of 200m with the same velocity of 25 m/s. When and where will the two bullets collide. ($g = 10\ m/s^2$)

NEET(UG) 2019 (Odisha) (+4 /-1)

1. after 2s at a height 180 m
2. after 2s at a height of 20 m
3. after 4 s at a height of 120 m
4. they will not collide

Question

A car starts from rest and accelerates at 5 m/s^2. At t = 4 s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at = 6 s ?

(Take g = 10 m/s^2)

NEET(UG) 2021 (+4 /-1)

1. 20 m/s, 5 m/s^2
2. 20 m/s, 0
3. $20\sqrt{2}$ m/s, 0
4. $20\sqrt{2}$ m/s, 10 m/s^2

Question

A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

If this particle were projected with the same speed at an angle ' θ ' to the horizontal, the maximum height attained by it equals 4R. The angle of projection, θ, is then given by: -

NEET(UG) 2021 (+4 /-1)

1. $\theta = \cos^{-1}\left(\frac{gT^2}{\pi^2 R}\right)^{1/2}$
2. $\theta = \cos^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{1/2}$
3. $\theta = \sin^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{1/2}$
4. $\theta = \sin^{-1}\left(\frac{2gT^2}{\pi^2 R}\right)^{1/2}$

Question

In the graph shown in fig. time is plotted along x-axis. Which quantity associated with a projectile motion is plotted along the y - axis?

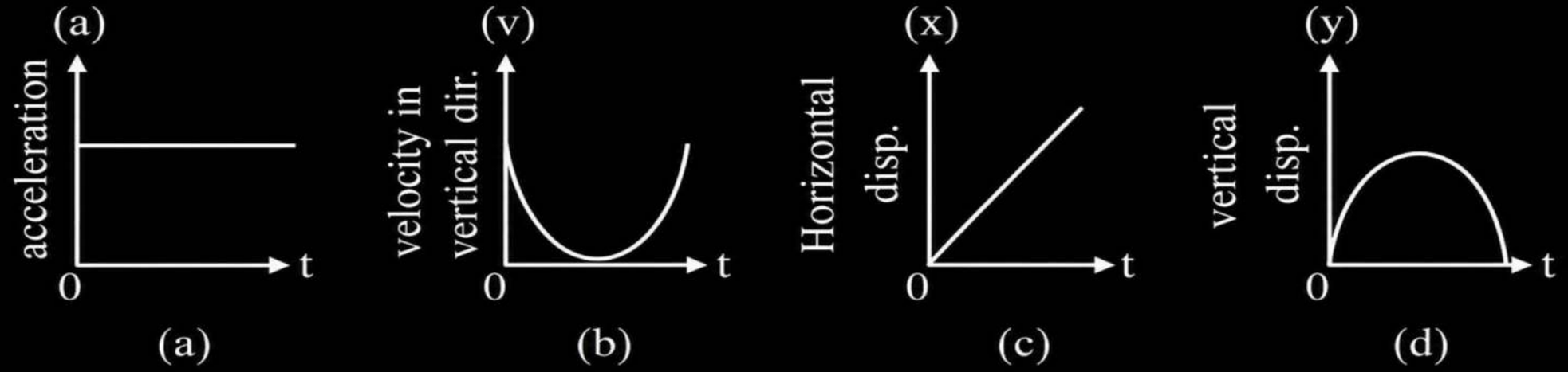


1. kinetic energy
2. momentum
3. horizontal velocity
4. none of the above

Question

For ground to ground projection following curves are given :-

Find incorrect one.



1. a
2. b
3. c
4. d

Question

Which of the following quantities remains constant during projectile motion?

(A) Average velocity between two points. (B) Average speed between two points

(C) $\frac{d}{dt}$ (D) $\frac{d^2}{dt^2}$ (?,

1. only A
2. only C
3. A and C
4. C and D

Question

In a projectile motion, the velocity :-

1. is always perpendicular to the acceleration
2. is never perpendicular to the acceleration
3. is perpendicular to the acceleration for one instant only
4. is perpendicular to the acceleration for two instants

Question

A body of mass m is thrown upwards at an angle θ with the horizontal with speed v. While rising up the magnitude of the velocity after t seconds will be:

1. $\sqrt{(v\cos\theta)^2 + (v\sin\theta)^2}$
2. $\sqrt{(v\cos\theta - v\sin\theta)^2 - gt}$
3. $\sqrt{v^2 + g^2t^2 - (2v\sin\theta)gt}$
4. $\sqrt{v^2 + g^2t^2 - (2v\cos\theta)gt}$

Question

A particle is fired with velocity u making θ angle with the horizontal. What is the magnitude of change in velocity when it returns to the ground.

1. $u\cos\theta$
2. u
3. $2u\sin\theta$
4. $(u\cos\theta - u)$

Question

In the above, the change in speed is :

1. $u\cos\theta$
2. 0
3. $u\sin\theta$
4. $(u\cos\theta - u)$

Question

The angle which the velocity vector of a projectile, will make with the vertical after time t of its being thrown with a velocity **v** at an angle $\boldsymbol{\theta}$ to the horizontal, is :

1. θ
2. $\tan^{-1}\left(\frac{\theta}{t}\right)$
3. $\tan^{-1}\left(\frac{v\cos\theta}{v\sin\theta - gt}\right)$
4. $\tan^{-1}\left(\frac{v\sin\theta - gt}{v\cos\theta}\right)$

Question

At a height 0.4 m from the ground, the velocity of a projectile in vector form is : Z . 2j) m / s . The angle of projection from horizontal is $(\mathbf{g = 10\ m/s^2})$: –

1. $45°$
2. $60°$
3. $30°$
4. $\tan^{-1}(3/4)$

Question

A particle is projected with a velocity u making an angle θ with the horizontal. At any instant, its velocity v is at right angle to its initial velocity u; then v is:

1. $u \cos \theta$
2. $u \tan \theta$
3. $u \cot \theta$
4. $u \sec \theta$

Question

A ball is projected upwards from the top of a tower with a velocity of 50 m/s making an angle of 30° with the horizontal. The height of the tower is 70 m. After how much time from the instant of throwing, will the ball reach the ground?

1. 2 s
2. 5 s
3. 7 s
4. 9 s

Question

Ball is projected with 30 m/s in horizontal direction from some height. Find time when it is 45° from horizontal.

General 2-D Motion

Must do ael Question

Question

x and y-coordinates of the particle at any time are $x = 5t - 2t^2$ and $y = 10t$, where x and y. Acceleration of the particle at $t = 2s$. **[NEET-2017]**

Question

A particle has initial velocity $2\hat{\imath} + 3\hat{\jmath}$ and acceleration $(0.3\hat{\imath} + 0.2\hat{\jmath})$. Magnitude of velocity after 10 sec. **[NEET-2012]**

Question

A position vector of a particle $r = 15t^2\,\hat{i} + (4 - 20t^2\,\hat{j})$. Find acceleration at $t = 1$ sec

[JEE Main 2019]

1. 25
2. 40
3. 100
4. 50

Question

Object starts from the point $(2\hat{\imath}, 4\hat{\jmath})$ m at $t = 0$ with velocity $(5\hat{\imath} + 4\hat{\jmath})$ with constant acceleration $(4\hat{\imath} + 4\hat{\jmath})$ m/s^2. What is distance from particle from origin at t = 2 sec.

[JEE Main-2019]

Question

Potion of object $\vec{r} = (t^2 - 38t)\,\hat{\imath} + 2t^3\,\hat{\jmath}$ find instant when velocity and acceleration are perpendicular. **[JEE-2022]**

Question

A particle starts from origin with velocity $3\hat{j}$ m/s and acceleration $(6\hat{i} + 4\hat{j})$. Then find x-coordinates of particle when y-coordinates is 32. **[IIT-2021]**

Question

Object is moving with velocity $V = 3 \sin(\omega t)\,\hat{i} + 3 \cos(\omega t)\,\hat{j}$ then find distance moved by object in 2 sec.

Question

Object is moving in west with 5 m/s after 2 sec its velocity is 5 m/s in north then find acceleration.

Thank

You